

# Multi-Modal EEG Feature Fusion for Upper-Limb Motor State Classification: A Comparative Deep Learning Study

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## Abstract

We compare classical Common Spatial Patterns with linear discrimination (Block I) against convolutional EEG decoders (Block II), including an EEGNet-style baseline and dual-branch models that fuse raw epochs with Welch spectra or wavelet coherence [1, 2]. All setups share pooled-stratified train-validation partitions on cue-locked upper-limb EEG. Quantitative scores and visuals are in Sections 3–4. Code: <https://github.com/MNR-MarcosOriolPagonabarraga-projects/HMI-EEGMovementClassification.git>

**Index Terms**—EEG motor decoding; common spatial patterns; deep learning; multimodal fusion; upper-limb classification.

## 1 Introduction

Decoding multiple upper-limb movement states from scalp EEG motivates brain-computer interfaces, neurorehabilitation, and robotic assistive devices [1]. Signal quality at the scalp is limited by spatial mixing (volume conduction), low SNR, and variability between trials, which complicates multiclass decoding.

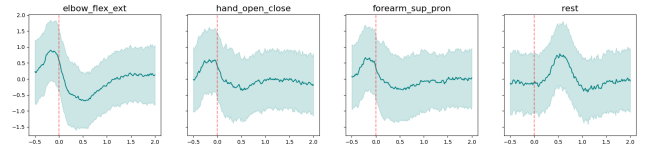
**Purpose of this work.** We organise the approach as two blocks applied to identical cue-locked data. **Block I** is a classical *CSP+LDA pipeline*: handcrafted spatial filters optimised for discriminative variance, followed by a linear Gaussian classifier [3].

**Block II** is a set of *convolutional decoders*: an EEGNet-style architecture [2] operating on raw voltage epochs together with two custom dual-branch models that concatenate or softly gate auxiliary tensors derived from Welch spectral maps and pairwise wavelet cross-spectral features.

**Contribution.** The comparison isolates where accuracy gains originate when moving from linear spatial prototypes to differentiable temporal receptive fields and, subsequently, multimodal fusion. Section 2 lays out methodology in that order (CSP, then convolutional architectures); Section 3 reports withheld accuracy and accompanying figures; Section 4 interprets them.

**Data.** Experiments reuse the Graz movement-execution corpus [1]. Code and preprocessing defaults are documented here:

<https://github.com/MNR-MarcosOriolPagonabarraga-projects/HMI-EEGMovementClassification.git>



**Figure 1:** Grand-average cue-locked EEG for four behavioural classes.

Figure 1 shows condition-averaged sensorimotor waveforms obtained after filtering and segmentation.

## 2 Materials and methods

### 2.1 Dataset and common preprocessing

We analysed openly released movement-execution EEG [1] recorded while volunteers sustained upper-limb actions with elbow, forearm, hand synergies, and resting fixation (BNCI Horizon 2020, accession 001-2017; 61 electrodes at 512 Hz [1]).

Experiments retained twenty-one peri-central electrodes, resampled the continuous signal at 128 Hz, and extracted cues locked to  $[-0.5, 2.0]$  s relative to the stimulus. Trials with peak EEG magnitude outside  $\pm 200 \mu\text{V}$  prior to augmentation were discarded. Mirror movement directions (flexion/extension, supination/pronation, open/close) were pooled, yielding four classes: *elbow*, *hand*, *forearm*, and *rest*.

All trainable architectures shared one stratified random partition of pooled subjects:  $\sim 85\%$  of trials optimisation,  $\sim 15\%$  validation, fixed pseudo-random seed for comparable convergence noise.

**Auxiliary spectral and connectivity tensors.** After per-trial, per-electrode temporal nor-

malization, Welch spectra between 3 and 30 Hz constitute log-magnitude PSD maps. Connectivity descriptors arise from modulus of pairwise complex Morlet cross-spectral densities at 10 and 22 Hz summed over unordered electrode pairs (then globally standardised). These tensors feed only the multimodal convolutional hybrids.

**Broadband filtering.** Recording segments were mains-notch filtered followed by FIR band-pass filtering between 0.5 and 40 Hz (zero-phase MNE kernels). CNN exports omitted common-average re-referencing whereas supplementary CSP optimisation sweeps exercised alternative motor-band hypotheses; divergence between those variants is flagged in Discussion.

## 2.2 Block I — CSP with LDA

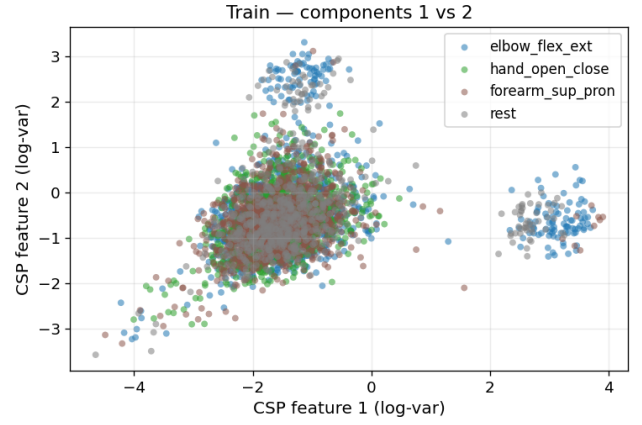
Block I implements the canonical motor-BCI blueprint: CSP spatial filters optimised for logarithmic-band-power variance ratios and immediately followed by linear discriminant analysis [3]. Six orthogonal spatial patterns provided feature vectors  $\mathbb{R}^6$  for LDA.

## 2.3 Block II — convolutional EEG networks

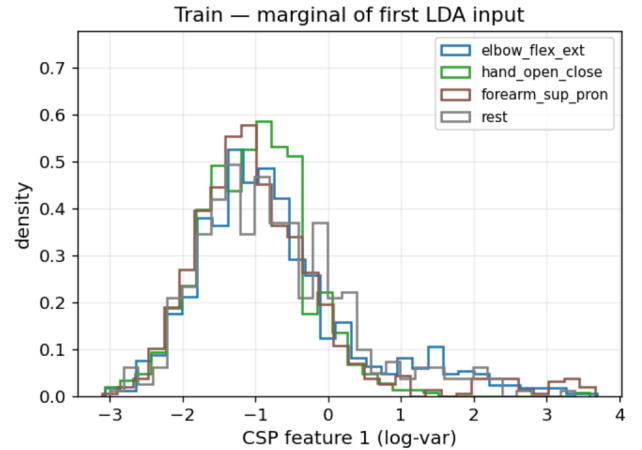
Block II instantiates differentiable temporal receptive fields analogous to EEGNet [2]:

1. **Baseline CNN** — separable convolutional stem processing raw EEG tensors [2].
2. **Spectral concat fusion** (EEGPSNet style) — late fusion by vector concatenation of CNN embeddings with shallow spectral CNN embeddings on Welch maps, batch-normalised prior to softmax.
3. **Connectivity attention fusion** (EEGConnectivityNet style) — independent encoders squeeze waveform embeddings and coherence tensors, softmax attention mixtures modulate modality weights, lastly MLP softmax classification.

All deep architectures were trained by minimising weighted cross-entropy with mild label smoothing, using adaptive moment (AdamW-type) updates with weight decay, reducing learning rates on withheld-loss plateaus, applying additive Gaussian EEG noise augmentation, and clipping gradients. Checkpoints retaining maximum withheld-set accuracy were kept for inference.



(a) CSP scattergram of logarithmic spectral energies (two projection axes illustrative).



(b) CSP marginal densities for exemplar discriminative component versus class histogram.

**Figure 2:** Ancillary exploratory plots for Block I feature geometry.

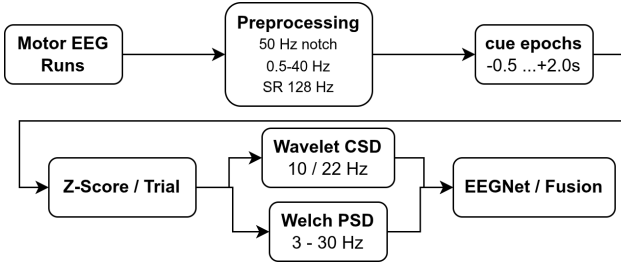
## 2.4 Ancillary visualisation protocols

Withheld softmax precursors underwent two-dimensional t-SNE with matched hyperparameters across Block II models. Connectivity scalp maps summarise trial-level coupling-derived quantities averaged separately for pooled hand epochs and pooled rest epochs prior to planar interpolation [1].

# 3 Experimental results

## 3.1 Stratification and accuracy benchmarks

Models were judged on withheld validation accuracy from the stratified pooled-subject split described in Section 2.1. CNN checkpoints correspond to maximal validation accuracy during optimisation. CSP+LDA occupies the pooled training partition for estimation and withheld validation for unbiased accuracy.



**Figure 3:** Simplified schematic of preprocessing branches feeding Block II deep models (PSD and connectivity paths optional depending on variant).

**Table 1:** Held-out multiclass classification accuracy.

| Approach series                          | Representations exploited                            | Held-out [%] |
|------------------------------------------|------------------------------------------------------|--------------|
| Block I CSP+LDA [3]                      | motor-band centred CSP (search over band hypotheses) | ~38–39       |
| Block I CSP+LDA [3]                      | broadband CSP (same 0.5–40 Hz envelope as Block II)  | ~32          |
| Block II CNN baseline [2]                | raw EEG temporal convolutions                        | ~56          |
| Block II CNN + PSD (EEGSPDNet-style)     | raw epochs + Welch maps                              | ~61          |
| Block II CNN + CSD attn. (CommNet-style) | raw epochs + coherence tensors                       | ~61–62       |

### 3.2 Connectivity scalp contrast maps

### 3.3 *t*-Distributed stochastic neighbour embedding of softmax precursors

### 3.4 Confusion matrices

## 4 Discussion

**Block I versus Block II.** Table 1 reports narrow-band exploratory CSP near  $\sim 38\text{--}39\%$  withheld accuracy, broadband CSP on tensors sharing the  $\sim 0.5\text{--}40$  Hz envelope with convolutional preprocessing at  $\sim 32\%$ , and Block II models on identical splits [2,3]. Convolutional receptive fields improve by  $\sim 17\text{--}18$  percentage points versus the optimised shallow CSP line and  $\sim 24$  versus broadband CSP paired to identical envelopes. This pattern suggests discriminative cues that exceed short CSP log-variance summaries of surface waveforms [3].

**CSP preprocessing parity.** The CSP rows entail different contracts: exploratory fits concentrate variance inside motor-frequency hypotheses, whereas broadband CSP aligns band limiting with Block II. Separately, tensor preparation here omitted common-average referencing for convolutional pipelines [3]. Fair CSP–CNN benchmarking must therefore unify band-pass envelopes, referencing, and cropping before blaming spatial filtering deficits alone.

**Multimodal fusion.** Welch concatenation and coherence attention each add roughly five percentage points over the EEGNet-style baseline [2]. Auxiliary maps likely supply marginal complementary constraints rather than orthogonal geometry.

**Neurophysiological structure versus accuracy.** Figure 4 exhibits organised peri-central con-

trast between trial-averaged hand-related epochs and rest [1] while Table 1 shows only modest incremental gains from fusion. Interpretable connectivity structure and softmax margins are therefore not interchangeable metrics.

**Embeddings and residual errors.** Colour intermixing in Fig. 5 parallels the accuracy plateau: softmax precursors from different classes remain neighbour points for every Block II model. Confusion matrices (Fig. 6) sharpen diagonals with depth, with remaining errors concentrated among kinematically related synergies.

**Generalisation scope.** Withholding trials from a pooled subject mix is optimistic relative to leave-one-subject-out evaluation recommended for neurotechnology translation [1].

## 5 Conclusion

We contrasted Block I CSP+LDA [3] with Block II convolutional decoders [2] using identical pooled-stratified withheld partitions [1].

Table 1 summarises withheld accuracies; interpretation of embeddings, topographies, fusion limits, and generalisation occupies Section 4.

Recommended follow-ups are aligned preprocessing for classical and deep pipelines, strictly subject-independent benchmarks [1], and fusion designs that exploit auxiliary tensors without weakening temporal discriminability.

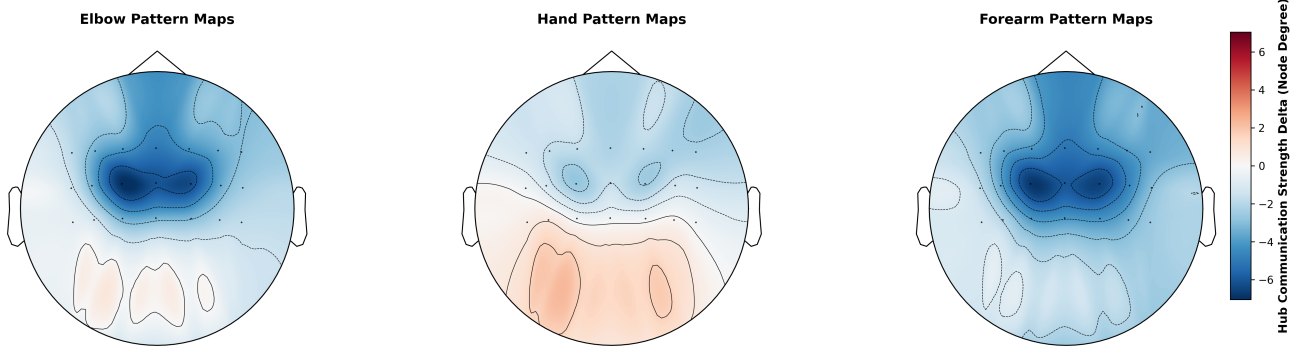
## Acknowledgment

Companion code:  
<https://github.com/MNR-MarcosOriolPagonabarraga-projects/HMI-EEGMovementClassification.git>

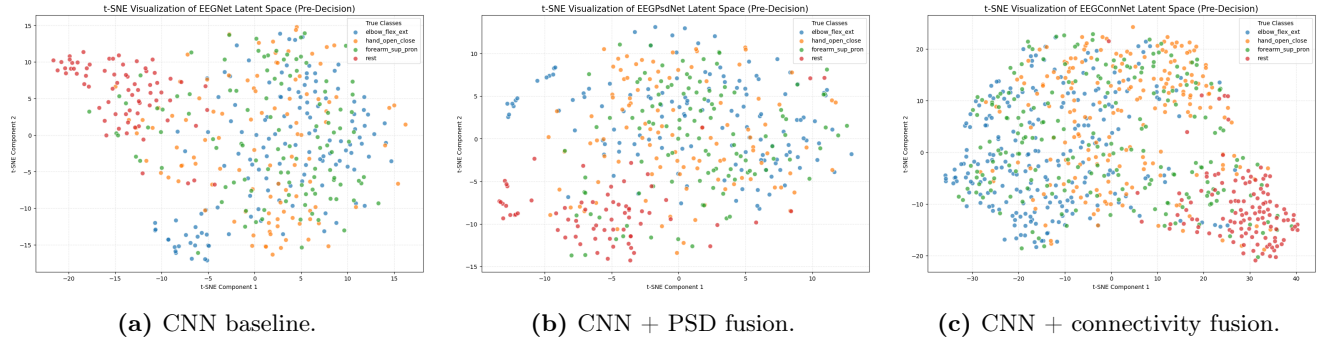
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- [2] V. J. Lawhern, A. J. Solon, N. R. Waytowich, S. M. Gordon, C. P. Hung, and B. J. Lance, “EEGNet: A compact convolutional neural network for EEG-based brain–computer interfaces,” *Journal of Neural Engineering*, vol. 15, no. 5, p. 056013, 2018.

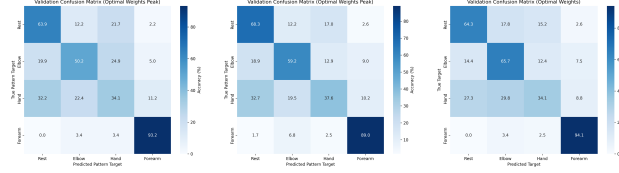
### Beta Band (22Hz) Functional Hub Activation Zones: Movement minus Rest



**Figure 4:** Trial-averaged connectivity-derived spatial patterns contrasting hand synergy epochs versus rest (Graz layout [1]).



**Figure 5:** Two-dimensional t-SNE embeddings of withheld-trial softmax input vectors.



**Figure 6:** Row-normalised confusion matrices (percent predicted per conditioned row) evaluated at withheld-set accuracy peaks for each Block II architecture.

- [3] A. Gramfort, M. Luessi, E. Larson, D. A. Engemann, C. Brodbeck, R. Goj, M. Jas, T. Brooks, L. Parkkonen, and M. S. Hämäläinen, “MEG and EEG data analysis with MNE-Python,” *Frontiers in Neuroscience*, vol. 7, p. 267, 2013.